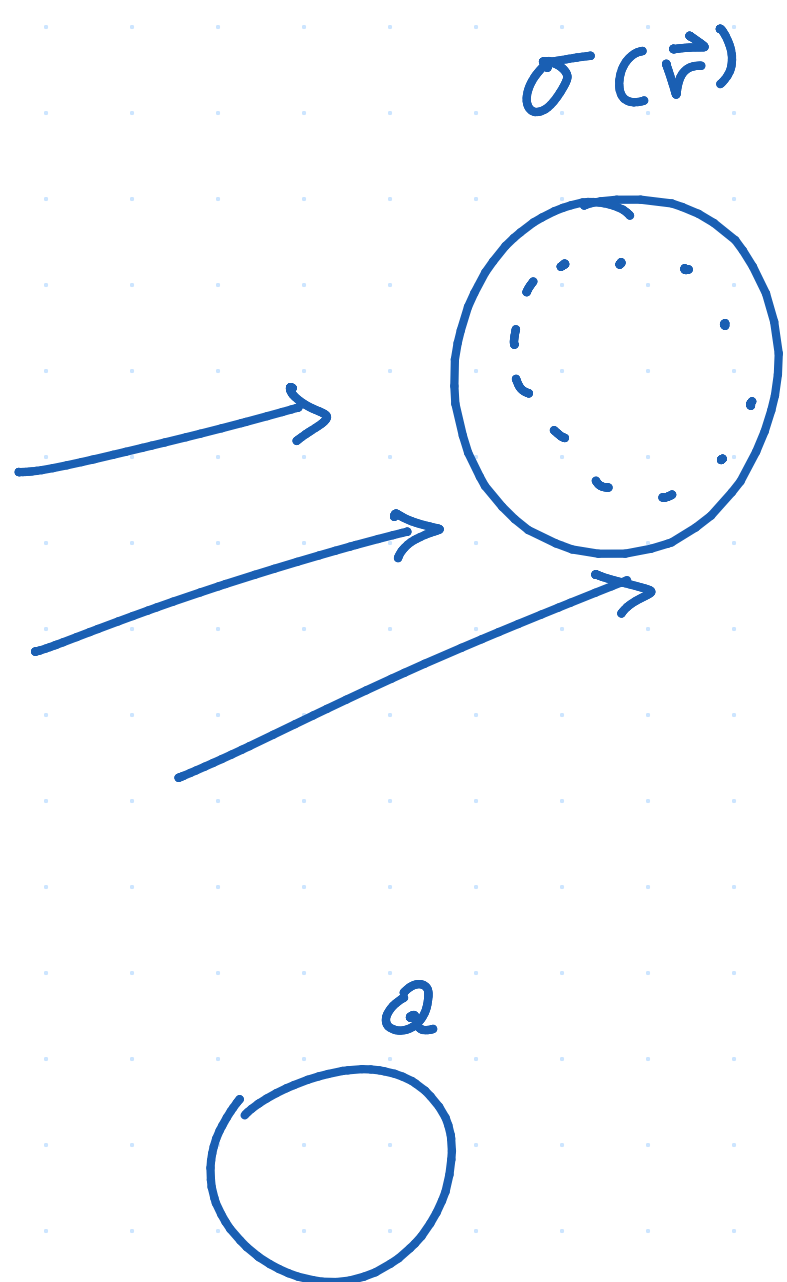


29/08/2025

Class #13

Electrodinámica

Unit 1 : Electrostatics



Poisson equation

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}') dV'}{(\vec{r} - \vec{r}')}^2$$

¿Como surge la ecuación de Poisson?

$\vec{E} = -\nabla U(\vec{r})$ ✓

$$\oint_S \vec{E} \cdot d\vec{a} = \int_{V'} \nabla \cdot \vec{E} dV' = \int_{V'} \frac{\rho(\vec{r}) dV'}{\epsilon_0}$$

✓✓ $\nabla \cdot \vec{E} = \frac{\rho(\vec{r})}{\epsilon_0}$ Gauss' law

Tenemos

$$\nabla \cdot \vec{E} = -\nabla \cdot \nabla U = \frac{\rho(\vec{r})}{\epsilon_0}$$

$$\Rightarrow \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = -\frac{\rho(\vec{r})}{\epsilon_0}$$

Laplacian Operator } $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

it is,

$$\nabla^2 U = -\frac{\rho}{\epsilon_0}$$

Ecuación de Poisson

$$\nabla^2 U = 0$$

Ecuación de Laplace

Laplaciano en Cartesianas

$$\nabla^2 U = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2}$$

Esféricas:

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

Cilindricas

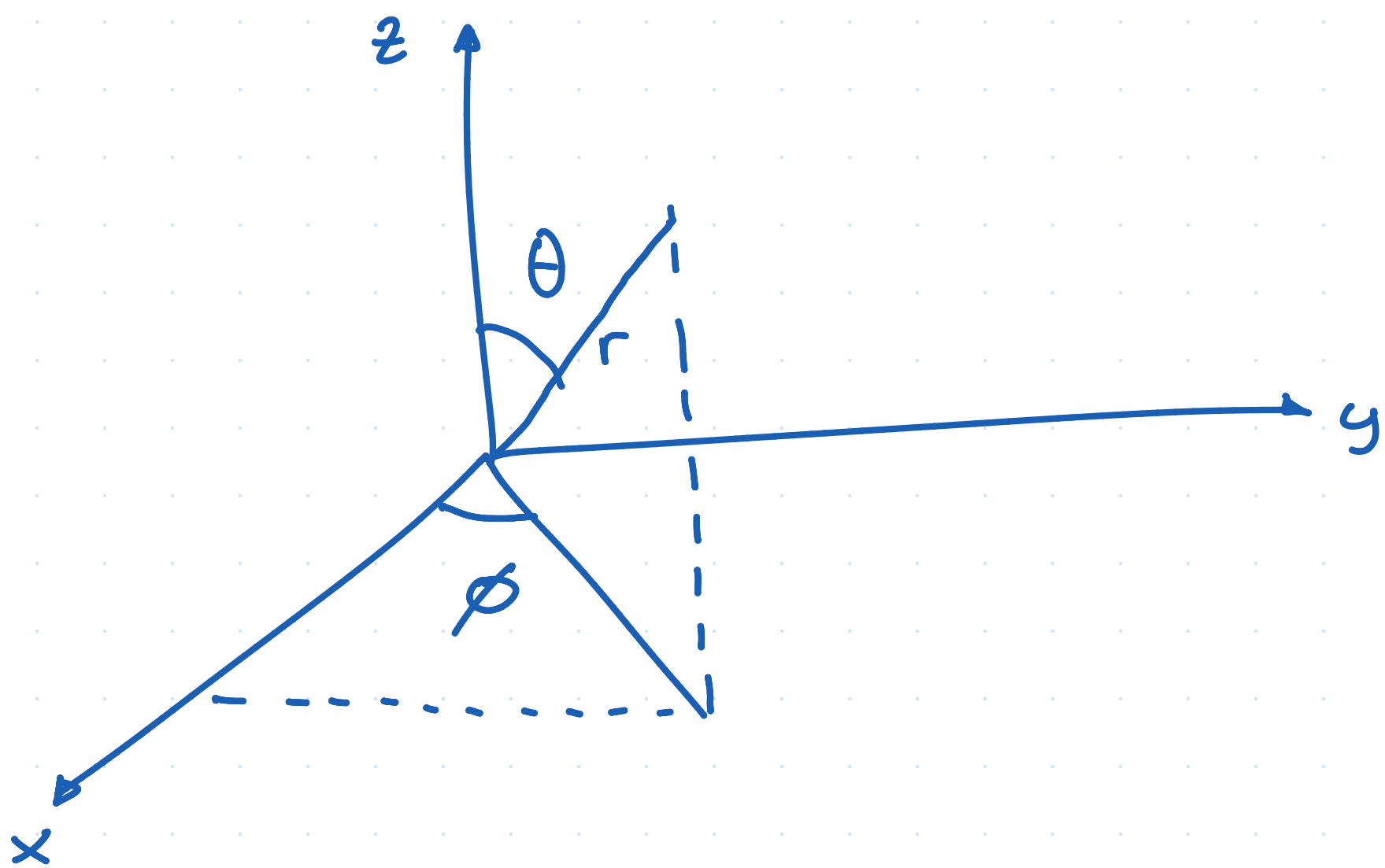
$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2}$$

The Poisson equation is defined independently of the coordinate system.

This equation in spherical coordinates is given by:

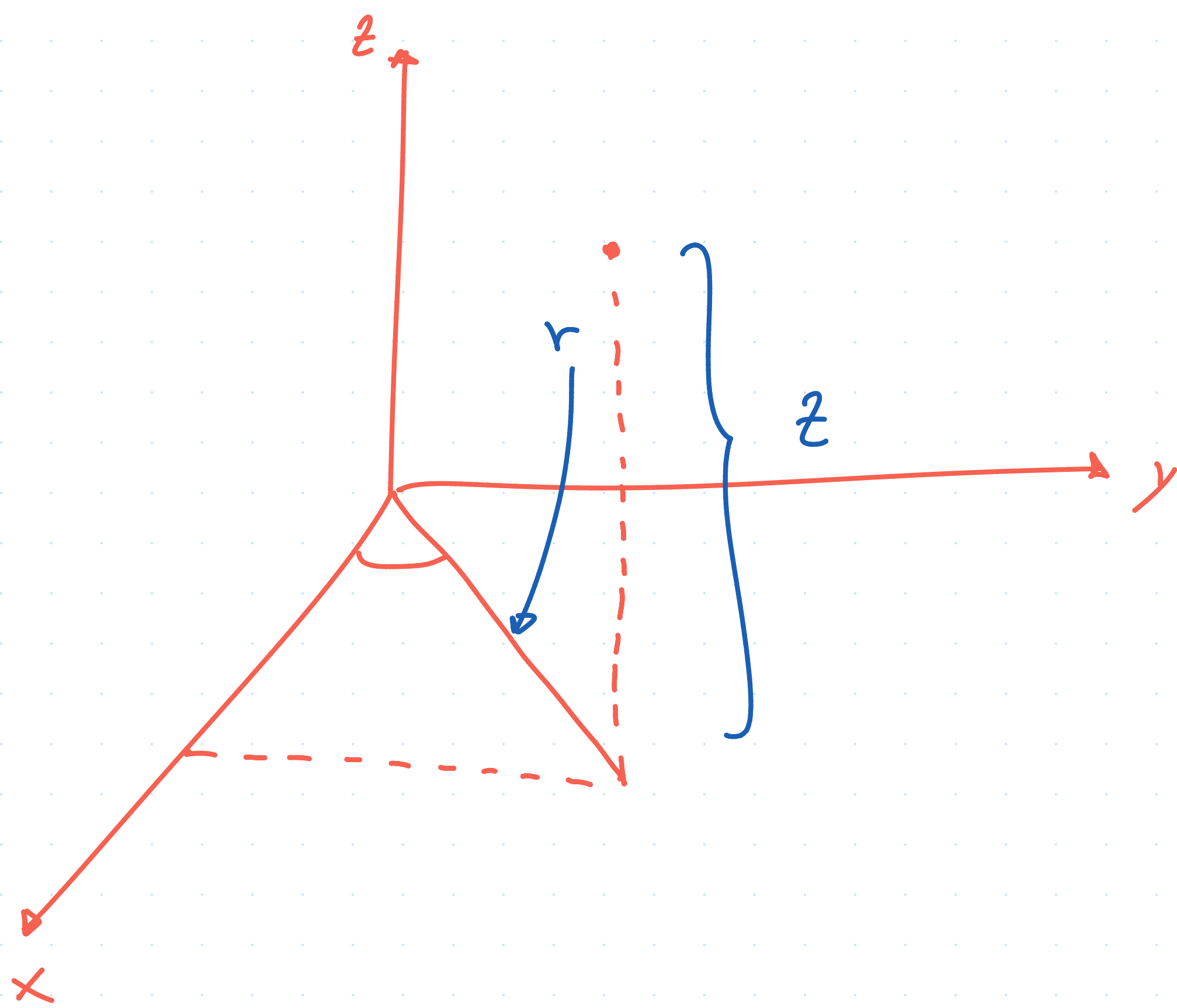
$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2} \left(\frac{1}{\sin^2 \theta} \right) \frac{\partial^2 u}{\partial \phi^2}$$

$$= - \frac{\rho(r, \theta, \phi)}{\epsilon_0}$$



In cylindrical coordinates;

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} = - \frac{\rho(r, \theta, z)}{\epsilon_0}$$



Laplace equation

If $\rho(\vec{r}) = 0$ in a volume V , then

$$\nabla^2 u = 0$$

Laplace equation

Considering an array of conductors with surfaces $S_1, S_2, \dots, S_n$

(with superficial charge), the Laplace equation is satisfied

in the interior of conductors

Theorem 1: If u_i ($i=1, n$) is solution to Laplace equation,

the $U = \sum_{i=1}^n C_i u_i$ is also a solution.

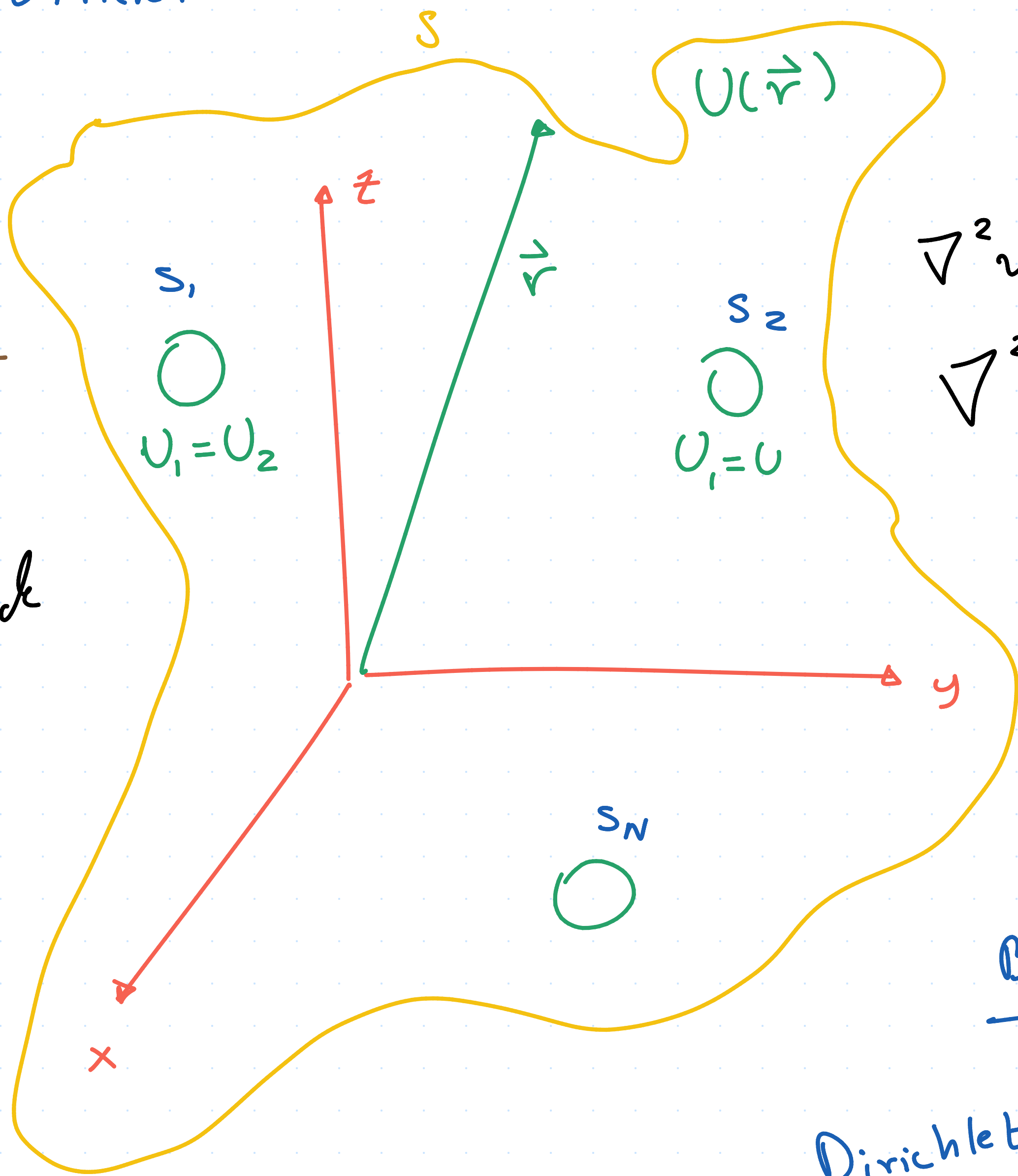
Teorema II | Interpretación:

↳ "Teorema de unicidad. Dos soluciones de la ecuación de Laplace que satisfacen las mismas condiciones en la frontera difieren a lo sumo en una constante aditiva"

condiciones de frontera:

↳ condiciones de Dirichlet
Neumann

de la frontera para el potencial electrostático



$U(\vec{r})$

$$\nabla^2 u_1(\vec{r}) = 0$$

$$\nabla^2 U_2(\vec{r}) = 0$$

$$U_1 = U_2 + C$$

Boundary condition

Dirichlet: U is specified on the boundary

• Neumann: $\frac{\partial U}{\partial n}$ is

specification on the boundary.

Theorem 2: consider two functions U_1 and U_2 satisfying the Laplace

equation in the volume V .

$$\nabla^2 (U_1 - U_2) = 0 \quad ; \quad \Phi = U_1 - U_2$$

U_1 ; U_2 = valen exactamente lo mismo si hay condiciones en la frontera (Dirichlet)

$\hookrightarrow \frac{\partial U}{\partial n}$ es condición de Neumann

• En condición de frontera para Dirichlet las funciones escalas son iguales.

$$\int_V \nabla \cdot (\phi \nabla \phi) = \oint_{S+S_1+S_2+\dots+S_N} \phi \nabla \phi \cdot d\vec{a}$$

Nota

$$\begin{aligned} \nabla \cdot (\phi \nabla \phi) &= \partial_i \phi \partial_i \phi \\ &\vdots \\ i &= \overline{1, 3} \end{aligned}$$

convención suma de Einstein:

$$\begin{aligned} &= (\partial_i \phi)(\partial_i \phi) + \phi \partial_i \partial_i \phi \\ &\vdots \\ &= \nabla \phi \cdot \nabla \phi + \phi \nabla^2 \phi \end{aligned}$$

$$0 = \int_V \nabla \cdot (\phi \nabla \phi) = \int_V \left[(\nabla \phi)^2 + \phi \nabla^2 \phi \right] dV$$

$$\Rightarrow \int_V (\nabla \phi)^2 = 0$$

Neuman : $\nabla \phi = 0$; $\nabla \phi \cdot \hat{n} = 0$ on the boundary $\frac{\partial u_1}{\partial n} = \frac{\partial u_2}{\partial n}$

In the volume

$$\nabla \phi \cdot \hat{n} = \frac{\partial \phi}{\partial n} = 0$$

$$\phi = \text{cte}$$

$$u_1 = u_2 + \text{cte}$$

Dirichlet : $\phi = 0$ on the boundary $\nabla \phi = 0$ means that

$$\phi = \text{cte} \Rightarrow u_1 = u_2 \checkmark$$